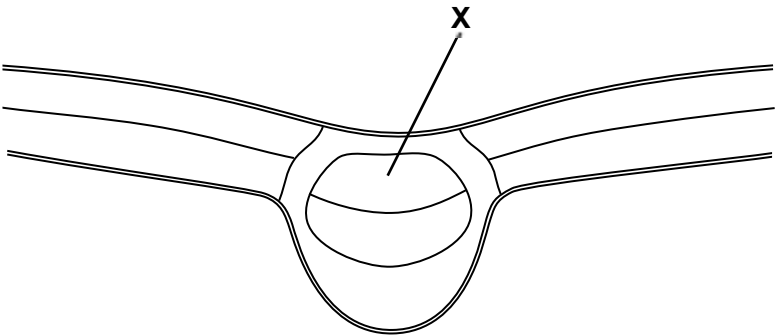


5(a)(i)	<i>if one phase not attempted, max 3</i> any four from: <i>S phase:</i> 1 DNA replication ; explanation: 2 produces two genetically identical, DNA molecules (one for each daughter cell ; A (sister) chromatids for DNA molecules A description of semi-conservative replication <i>metaphase:</i> 3 chromosomes / pairs of sister chromatids align / orientate at (spindle) equator ; A metaphase plate for equator ; <i>explanation:</i> 4 chromosomes orientated so that sister chromatids will be shared out to daughter cells / AW ; <i>anaphase:</i> 5 centromere splits / spindle fibres shorten / AW ; <i>explanation:</i> 6 so that daughter chromosomes / sister chromatids / identical chromatids, move to opposite poles ; 7 AVP ; e.g. ref. to checkpoints and ref. to error, detection / repair ref. to kinetochore / centromere, and attachment to spindle fibres at, metaphase / anaphase	4
5(a)(ii)	unequal sharing out of, cytoplasm / organelles / named organelles, (at cytokinesis) ; A uneven sizes	1

Question	Answer	Marks
5(b)(i)	<i>max 3 structural changes:</i> 1 ref. to function of xylem ; e.g. transport of water and (dissolved) mineral ions transport of water from roots to other parts of the plant <i>in context of leaf xylem</i> receipt of water from roots 2 end walls broken down ; 3 (so) tubes form (to transport water long distances) ; 4 (so), uninterrupted / unhindered / continuous, flow ; 5 ref lignification ; 6 lignin strengthens vessels ; 7 ref. to cell death ; 8 cells, become hollow / lose contents / lose cytoplasm or organelles / named organelles, digested / broken down ; 9 greater quantity of water can flow (per unit, time / length) ; 10 AVP ; e.g. tonoplast / vacuole, breakdown / rupture, to release hydrolytic enzymes ref. to xylem patterning during growth ref, to pits	4
5(b)(ii)	label line to xylem, labelled X ; 	1